Workshop - Week 10

1. Make a substitution to express the integrand as a rational function and then evaluate the integral.

(a) $\int \frac{1}{\sqrt{x}-\sqrt[3]{x}} dx$ (*Hint:* Substitute $u = \sqrt[6]{x}$)

(b) $\int \frac{e^{2x}}{e^{2x}+3e^x+2} dx$

(c) $\int \frac{\sin\ x}{\cos^2\ x-3\cos\ x}$

(d) $\int \frac{dx}{1+e^x}$

2. (a) Use integration by parts to show that, for any positive integer n,

$$\int \frac{dx}{(x^2+a^2)^n} dx = \frac{x}{2a^2(n-1)(x^2+a^2)^{n-1}} + \frac{2n-3}{2a^2(n-1)} \int \frac{dx}{(x^2+a^2)^{n-1}}$$

2. use part (a) to evaluate

$$\int \frac{dx}{(x^2+1)^2} \text{ and } \int \frac{dx}{(x^2+1)^3}$$

3. Determine whether each integral is convergent or divergent. Evaluate those that are convergent

(a) $\int_{-\infty}^{\infty} xe^{-x^2} dx$

(b) $\int_1^{\infty} \frac{\ln\ x}{x} dx$

(c) $\int_1^{\infty} \frac{\ln\ x}{x^2} dx$

(d) $\int_e^{\infty} \frac{1}{x(\ln\ x)^2} dx$

(e) $\int_0^1 \frac{dx}{x}$